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Project Phase-1 Review-2

Project Title: ReconIQ: An Agentic Financial Reconciliation System

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Introduction

- Financial reconciliation is essential for ensuring consistency between bank statements, internal ledgers, and payment-gateway records.
- Transactions frequently appear with different descriptions, dates, and formats across financial systems.
- Traditional exact-match techniques struggle with complex settlements, gateway fees, and transactions represented by multiple ledger entries.
- AI can help evaluate ambiguous transaction candidates, but an AI-generated decision may still be incorrect.
- ReconIQ proposes a hybrid reconciliation system combining deterministic matching, AI-assisted reasoning, independent verification, and auditable results.

Problem Statement

- Financial data from different sources often contains inconsistent descriptions, formats, dates, and transaction identifiers.

- A single bank settlement may correspond to multiple internal ledger entries after payment-gateway fees.
- Similar transaction amounts and dates can create multiple plausible matching candidates.

- Simple rule-based systems may fail to identify complex relationships, while unrestricted AI matching risks accepting incorrect transactions.

- There is a need for a reliable reconciliation system that verifies proposed matches and leaves uncertain cases unresolved rather than guessing.

Literature Review

Study	Key Findings
Fellegi & Sunter (1969), A Theory for Record Linkage	Established a statistical framework for classifying records as matches, non-matches, or possible matches. Supports ReconiQ's conservative treatment of uncertain transactions.
Mudgal et al. (2018), Deep Learning for Entity Matching: A Design Space Exploration	Found that deep learning can outperform existing methods on textual and noisy records, but does not always improve structured-data matching. Supports combining deterministic rules with semantic evidence.
Christen (2012), A Survey of Indexing Techniques for Scalable Record Linkage and Deduplication	Examined techniques that reduce the number of record comparisons while maintaining matching quality. Highlights the importance of efficient candidate selection.
Du et al. (2024), Improving Factuality and Reasoning in Language Models through Multiagent Debate	Reported improvements in reasoning and factual validity through multi-agent debate. Provides research motivation for ReconiQ's Proposer-Verifier workflow.
Smit et al. (2024), Should We Be Going MAD? A Look at Multi-Agent Debate Strategies for LLMs	Found that multi-agent debate does not reliably outperform alternative prompting methods. Highlights why ReconiQ's verification stage must be evaluated rather than assumed to improve accuracy.

Methodology

1. Data Collection

- Import bank statements, internal ledger records, and payment gateway reports.

2. Data Preprocessing

- Parse and standardize transaction fields.
- Validate transaction amounts, dates, and reference IDs.
- Identify missing or inconsistent information.

3. Deterministic Matching

- Match transactions using reference IDs and exact values.
- Apply date and amount tolerances where appropriate.

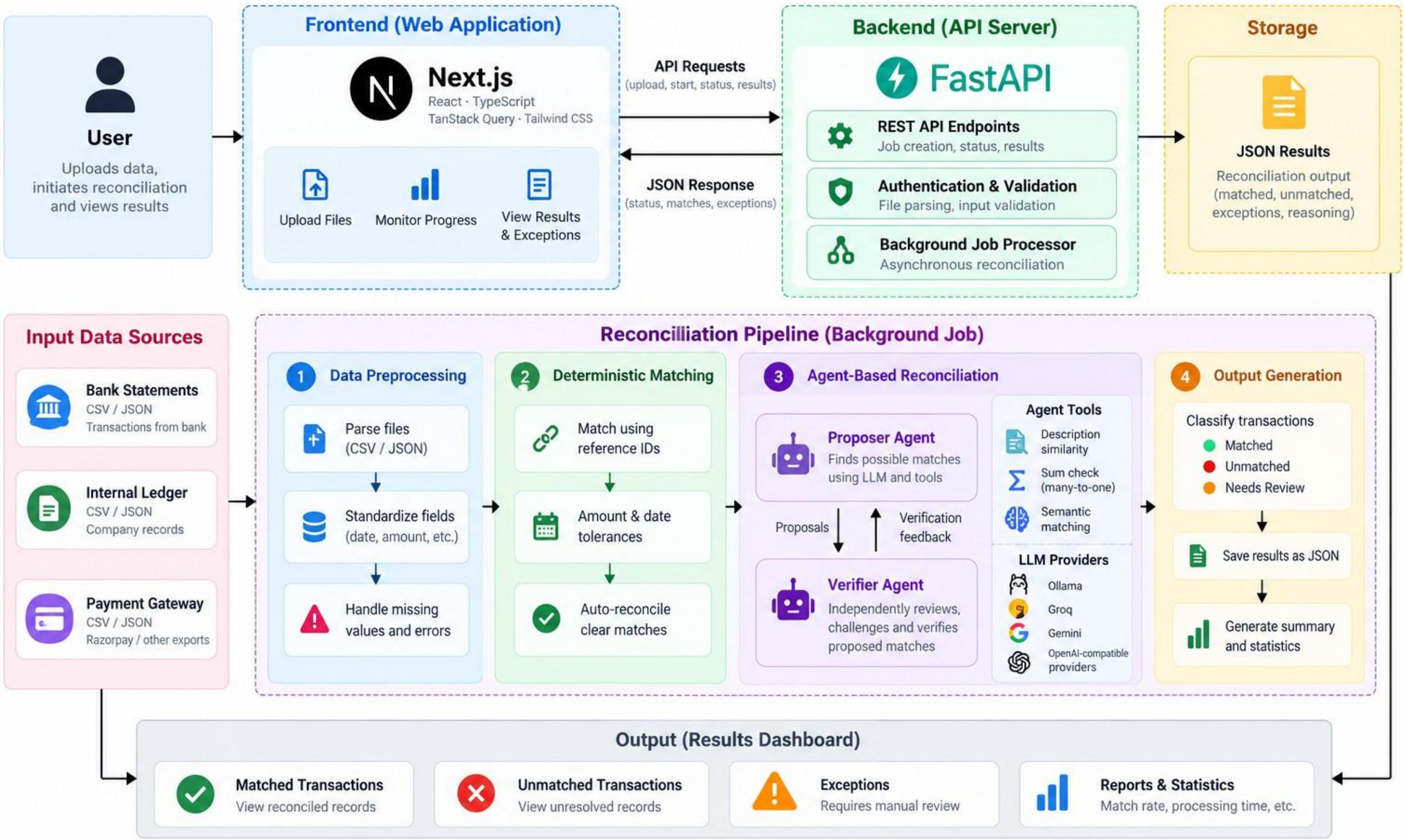
4. Agent-Based Reconciliation

- Send ambiguous transactions to the Proposer agent for candidate matching.
- Use tools such as description similarity and sum checks to investigate possible matches.
- Let the Verifier agent independently review proposed matches.

5. Evaluation and Exception Handling

- Classify transactions as matched, unmatched, or requiring review.
- Measure performance using precision, recall, and processing time.
- Present reconciliation results and unresolved exceptions through the interface.

System Architecture



Expected Results

- Accurate classification of financial transactions into matched, unmatched, and exception categories.
- Effective identification of complex transaction matches using the hybrid deterministic and AI-based approach.
- Improved reconciliation efficiency compared with conventional manual or rule-based methods.
- Reliable handling of ambiguous transactions through the Proposer-Verifier workflow.
- A foundation for automated financial reconciliation with transparent results and exception handling.

Evaluation Metrics

Area	Metric
Matching Performance	Precision and match rate
Classification Quality	Precision, Recall, F1-Score
Processing Efficiency	Reconciliation time and processing speed
Verification Quality	Correctly verified matches and unresolved exceptions
Comparative Performance	ReconIQ vs. deterministic-only matching

Conclusion

- ReconIQ automates financial reconciliation across bank statements, internal ledgers, and payment gateway records.
- Deterministic matching resolves clear transactions using reference IDs, amounts, and date tolerances.
- The Proposer-Verifier agent workflow investigates ambiguous transactions and verifies proposed matches.
- Supporting tools perform description similarity checks and many-to-one sum validation.
- Performance can be evaluated using precision, recall, F1-score, match rate, and processing time.
- ReconIQ provides a foundation for efficient, scalable, and transparent financial reconciliation with explicit exception handling.

References

1. Fellegi, I. P., & Sunter, A. B. (1969). “A Theory for Record Linkage.” Journal of the American Statistical Association, 64(328), 1183–1210. https://doi.org/10.1080/01621459.1969.10501049

2. Mudgal, S., et al. (2018). “Deep Learning for Entity Matching: A Design Space Exploration.” Proceedings of the 2018 International Conference on Management of Data (SIGMOD). https://doi.org/10.1145/3183713.3196926

3. Christen, P. (2012). A Survey of Indexing Techniques for Scalable Record Linkage and Deduplication. IEEE Transactions on Knowledge and Data Engineering. Google Scholar

4. Du, Y., et al. (2024). “Improving Factuality and Reasoning in Language Models through Multiagent Debate.” International Conference on Machine Learning (ICML). arXiv search

5. Smit, A., et al. (2024). “Should We Be Going MAD? A Look at Multi-Agent Debate Strategies for LLMs.” International Conference on Machine Learning (ICML). Google Scholar